SHREE VENKATESHWARA
HI-TECH ENGINEERING COLLEGE
(AUTONOMOUS)
GOBI-638 455

NAME OF THE LAB_____

University Register No_____

Name _____ Roll No_____

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Certified that this is bonafide record of work done by the above student during the year 20 -20

LAB IN-CHARGE **HEAD OF THE DEPARTMENT**

Submitted for the practical Examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

S. No.	NAME OF THE EXPERIMENT	PAGE No.	MARKS	SIGN
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PRACTICAL EXERCISES: 30 PERIODS

1. Install Kali or Backtrack Linux / Metasploitable/ Windows XP

2. Practice the basics of reconnaissance.

3. Using FOCA / SearchDiggity tools, extract metadata and expanding the target list.

4. Aggregates information from public databases using online free tools like Paterva's Maltego.

5. Information gathering using tools like Robtex.

6. Scan the target using tools like Nessus.

7. View and capture network traffic using Wireshark.

8. Automate dig for vulnerabilities and match exploits using Armitage

FOCA : http://www.informatica64.com/foca.aspx.

Nessus : http://www.tenable.com/products/nessus.

Wireshark : http://www.wireshark.org.

Armitage : http://www.fastandeasyhacking.com/.

Kali or Backtrack Linux, Metasploitable, Windows XP

Ex:No.: 1 Date:	**Install Kali or Backtrack Linux / Metasploitable/ Windows XP**

Aim:

To Install Kali Linux on Windows using Oracle Virtual Box

Procedure:

1. VirtualBox Installation:

- Proceed to download VirtualBox fromthe official website(https://www.virtualbox.org/wiki/Downloads)



2. Downloading Kali Linux ISO Image:

- Following VirtualBox setup, obtain the Kali Linux ISO image from the provided link(https://www.kali.org/get-kali/#kali-installer-images)
- You have the option to either directly download the ISO or employ the torrent for the download.



3. Create a new virtual machine and configure storage, memory:

- With VirtualBox successfully installed, commence the application via the start menu.
- Select the "New" option to initiate the creation of a fresh virtual machine.



- Name your virtual machine.
- Keep the default folder name for the VM file.
- Browse and select the Kali Linux ISO file.
- Set the Type to Linux and Version to any 64-bit Linux.
- Configure the RAM (recommend 2-4 GB for an 8 GB RAM system).
- Choose 2 or 4 processors.
- In the Hard Disk section, keep the file location as is.
- Allocate at least 20-25 GB for Kali Linux installation.
- Select VDI as the hard disk file type.
- Do not select pre-allocate full size.
- Click "Finish" to create the virtual machine.

4. Install Kali Linux on the virtual machine:

- With the virtual machine configured, you can now proceed with the installation of Kali Linux. To do this, click the "Start" button in the toolbar orright-click > selectstart > normalstart.



- After a successful boot, you should see a screen showing various options for installing Kali Linux.
- Select **the first default option** to start the installer.



- In the next few screens,select Language,region, and other basic details per your location.
- Add any hostage you want.



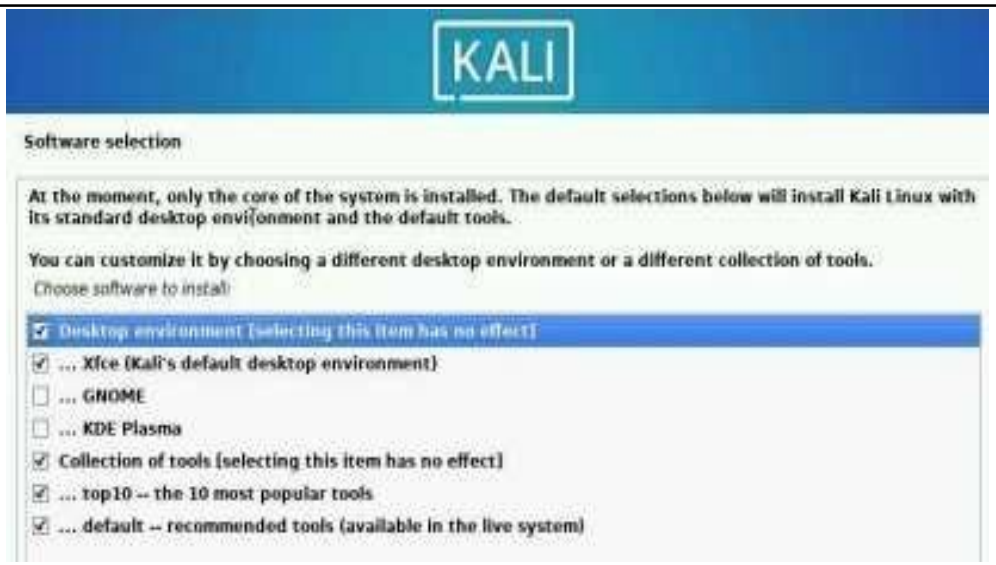
- In the nextscreen, keep the domain name blank. Then give your name and user name(theloginname). Enter the password for that user.
- In the partition screen,select "guided – use the entire disk". This is the best option for installing in VirtualBox. If installing it on the physical system, do not use this.



- Continue with the installation as per the on-screen instructions.



- When the following prompt appears,select the following ones(default), which feature the Xfce Desktop environment and key hacking tools. Hit enter to continue.



- Wait For The Installation to finish. Kali Linux installation VirtualBox will take a few minutes (approximately 10 minutes). While nearing the end of the installation,the installer will ask you about the GRUB installation. Answer Yes and choose the device /dev/sda and continue.



- Click on Restart Now after the installation is complete. Wait for a fewseconds and you should be here on the login screen.
- Use the user id and password to log in. And you should see KaliLinux desktop is running inside VirtualBox as VM in Windows.

Result:

The installation of Kali Linux on VirtualBox was successful, creating a safe space to explore ethical hacking tools and techniques.

Ex:No.: 2	**Practising the Basics of Reconnaissance in Ethical Hacking.**

Aim:

To Practise the Basics of Reconnaissance in Ethical Hacking.

Terminal Commands and Their Use:

1. **Ping Scan with Nmap:**
   - **Command:** nmap -sn <target_ip>
   - **Use:** Perform a ping scan to determine which hosts are up in a given IP range.

Sample Output:

```
Starting Nmap 7.91 ( https://nmap.org ) at 2023-08-09 10:00 EDT
Nmap scan report for <target_ip>
Host is up (0.034s latency).
MAC Address: XX:XX:XX:XX:XX:XX (Manufacturer)
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

2. **DNS Enumeration with Dig:**
   - **Command:** dig <target_domain>
   - **Use:** Perform DNS enumeration to retrieve information about a target domain's DNS records

Sample Output:

```
; <<>> DiG 9.16.1-Ubuntu <<>> <target_domain>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: XXXXX
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; QUESTION SECTION:
;<target_domain>.     IN  A

;; ANSWER SECTION:
<target_domain>.  300 IN  A   <target_ip>

<target_domain>.  300 IN  A   <another_ip>

;; Query time: 10 msec
;; SERVER: 127.0.0.53#53(127.0.0.53)
;; WHEN: Mon Aug 09 10:05:53 EDT 2023
;; MSG SIZE  rcvd: 76
```

3. WHOIS Lookup:

- **Command:** whois <target_domain>
- **Use:** Retrieve registration and ownership information about a domain.

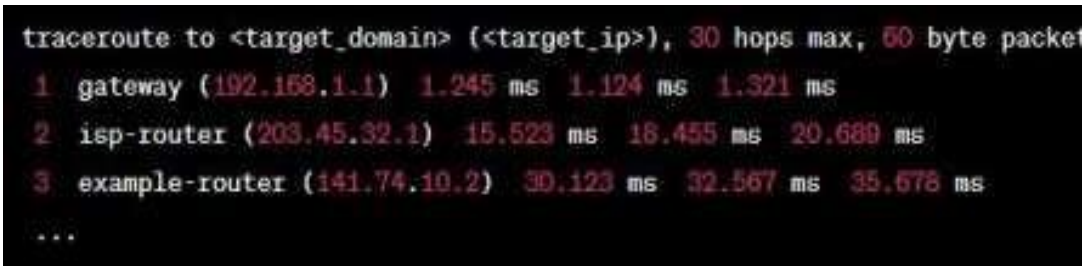
Sample Output:



4. Traceroute:

- **Command:** traceroute <target_domain>
- **Use:** Identify the path packets take from your system to the target, revealing intermediate routers' IP addresses.

Sample Output:



Result:

Thus, essential reconnaissance techniques in ethical hacking were practiced using terminal commands, successfully enabling the gathering of information about target IP addresses, domains, DNS records, and registration details.

Ex:No.: 3	Using FOCA / SearchDiggity tools, extract metadata and expand the target list.

Aim:

To understand how to extract metadata from a website using FOCA (Fingerprinting Organizations with Collected Archives) software.

Procedure:

1. Setting up SQL Server

- Open a web browser and navigate to the Microsoft SQL Server Express download page.
- Click on the 'Download' button to download the installer.
- Once downloaded, double-click on the installer file.
- Accept the terms and conditions.
- Select the installation type as "Basic" and proceed.
- The installer will take care of the rest. Wait for the installation to complete.

2. Installing FOCA Software

- Download FOCA from its official website or a trusted source.
- Once downloaded, locate the downloaded ZIP file and extract it.
- Open the extracted folder and double-click on FOCA.exe to launch the application.

3. Creating a New Project in FOCA

- Upon opening FOCA, click on 'New Project'.
- Name your project and in the 'Domain' field, enter the website from which you want to extract metadata.
- Click on 'Create Project'.

4. Searching and Downloading Documents

- Select the search engines you want FOCA to use.
- Choose the types of documents you want to search for (like PDF, DOC, PPT, etc.).
- Click on 'Search All'.
- A list of files related to your search will appear.
- Right-click on any listed file and choose 'Download All' to download all listed files.

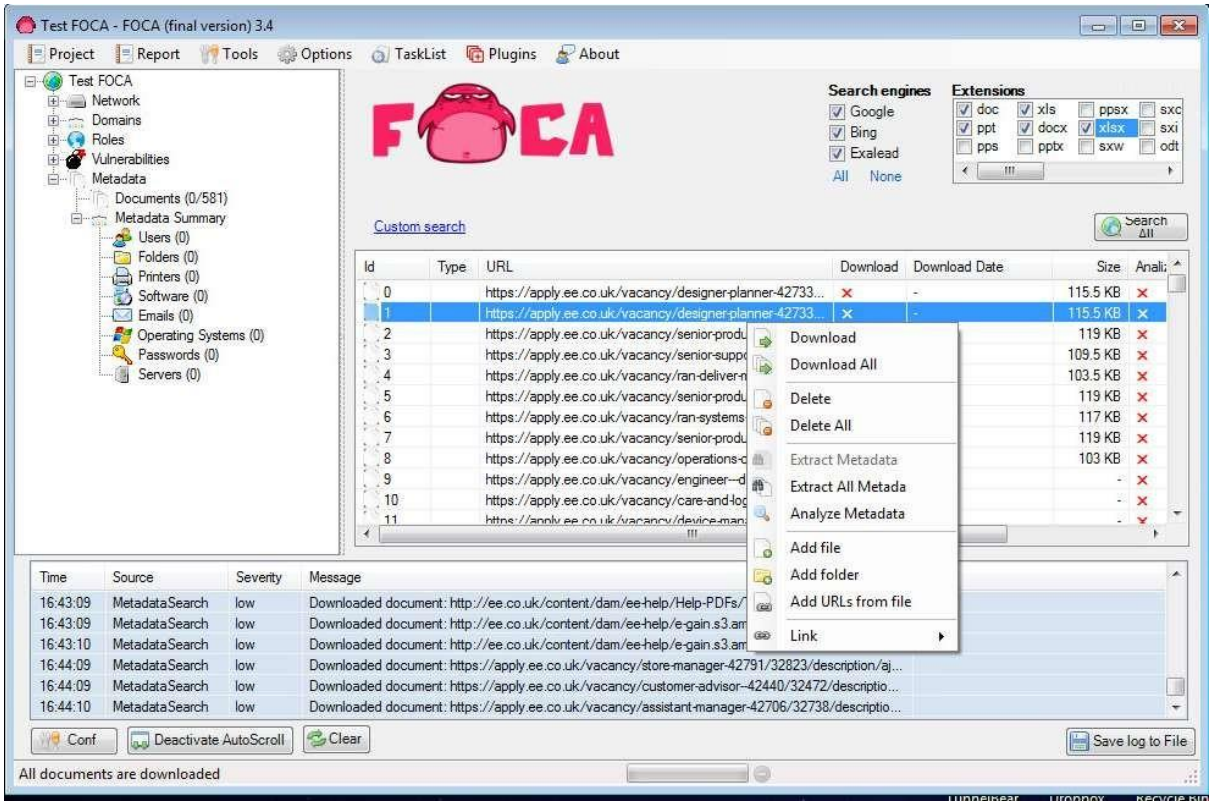
5. Extracting Metadata

- Once all files are downloaded, right-click again on any downloaded file in FOCA.
- Choose 'Extract All Metadata'.
- Wait for the process to complete.

6. Viewing Extracted Metadata

- On the left side of the FOCA interface, click on 'Metadata'.
- You will see a list of metadata information extracted from the downloaded files.

Output:



Result:

Thus, extraction of metadata from a website using FOCA (Fingerprinting Organizations with Collected Archives) software has been successfully done.

Ex:No.: 4	**Aggregates information from public databases using online free tools like Paterva‘s Maltego.**

Aim:

The aim of this lab is to teach you how to use Paterva's Maltego for aggregating information from public databases.

Procedure:

1. Installing Maltego

- Visit the official Maltego website to download the software. Make sure to choose the correct version for your operating system.
- Install Maltego by following the on-screen instructions.
- Once the installation is complete, open the Maltego application.
- Register or log in to your Maltego account.

2. Creating a New Graph

- Click on the 'File' menu in the Maltego interface and then select 'New'.
- A new graph window will open, which will be your workspace for this session.

3. Basic Entity Search

- Drag and drop an entity type (like "Domain", "Email Address", etc.) from the Entity Palette on the left into the graph.
- Double-click on the entity and enter the specific domain or email address you want to investigate.

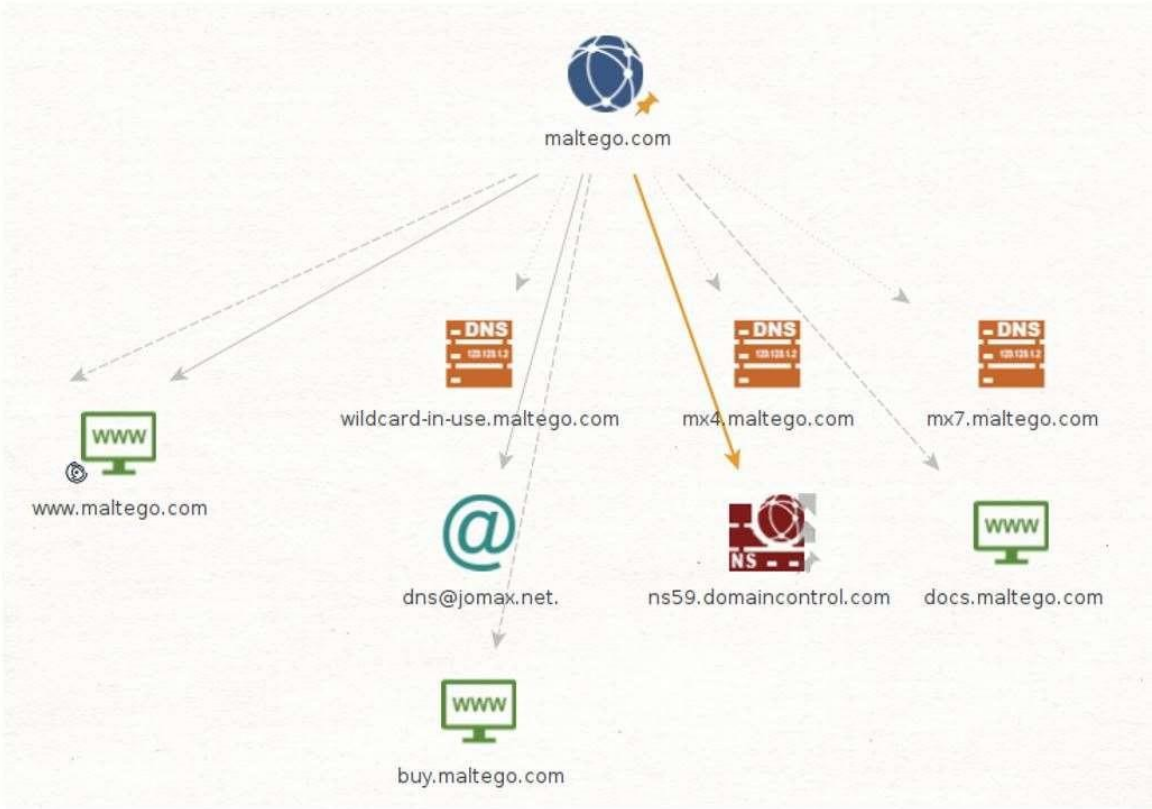
4. Run a Transform

- Right-click on the entity you've placed on the graph.
- From the 'Run Transform' menu, select an appropriate transform. For example, you might choose "To Email [Using Search Engine]" for an email entity.
- Maltego will execute the transform and provide you with new entities connected to your original entity.

5. Exploring and Aggregating Information

- Continue to run more transforms on the new entities that appear on your graph to gather more information.
- You can adjust your view, zoom in/out, and rearrange entities as needed for better understanding.
    - Aggregated data might include things like associated social media accounts,related websites, IP addresses, etc.

Output:



Result:

Thus, Information Aggregation with Paterva's Maltego has been successfully done and executed.

Ex:No.: 5	Information gathering using tools like Robtex.

Aim:

The aim of this lab exercise is to understand how to gather information on domains, IP addresses, and networks using Robtex, an online tool that provides various kinds of data for network research.

Procedure:

1. Accessing Robtex

- Open your web browser and go to the Robtex website by navigating to https://www.robtex.com/.
- You will see a search bar on the main page, where you can enter the domain name, IP address, or network that you wish to investigate.

2. Domain Lookup

- In the Robtex search bar, type the domain you want to investigate. For example, example.com.
- Click the 'Search' button or press Enter.
- Robtex will return a wealth of information about the domain, including DNS records, IP address data, server location, and more.

3. IP Address Lookup

- In the Robtex search bar, enter the IP address you want to investigate.
- Hit the 'Search' button or press Enter.
- Robtex will provide data related to the IP address, such as the owning organization, geolocation, ASN information, and potentially linked domains.

4. ASN Lookup

If you want to search for an Autonomous System Number (ASN), enter it into the search bar.
Press the 'Search' button.
Information like AS routes, IP ranges, and associated domains will be displayed.

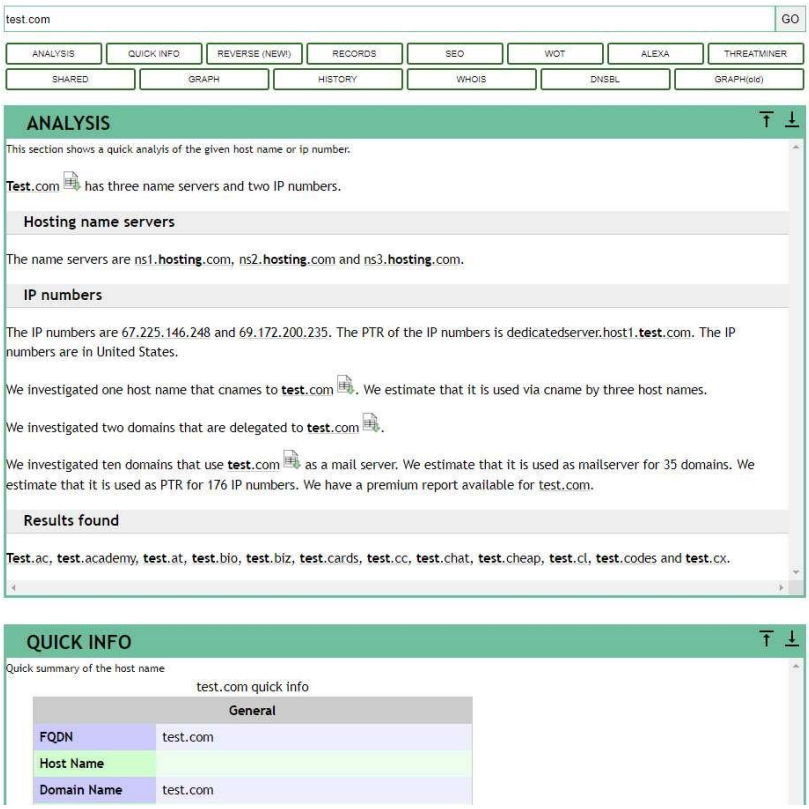
5. Understanding and Analyzing Data

- Take your time to go through the different tabs and sections that Robtex offers for each type of search. This can include but is not limited to:
    - DNS records
    - Mail server records

- Domain siblings and subdomains
- Shared IPs
- ASN information

- Each section can offer valuable insights into the domain, IP, or network you are investigating.

Output:



Result:

Thus, the gathering of information using Robtex has been successfully done and executed.

Ex:No.: 6	Scanning a Target Using Nessus

Aim:

To understand how to use Nessus for vulnerability scanning on a target system or network.

Procedure:

1. Initial Setup and Login

- Open Nessus by navigating to the application via your program menu or using the desktop shortcut if available.
- If you're using Nessus for the first time, you will need to activate it. You can get an activation code by registering on the Tenable website.
- Once activated, log in to the Nessus interface using your credentials.

2. Updating Plugins

- Before starting your scan, it's recommended to update Nessus plugins to get the latest vulnerability checks.
- Usually, Nessus updates its plugins automatically, but you can manually update them by navigating to the 'Plugins' section and clicking 'Update'.

3. Creating a New Scan

- Once logged in, click on the 'New Scan' button on the dashboard.
- You will be presented with various scanning templates like Basic Network Scan, Advanced Scan, etc. Choose the one that suits your needs.
- Name your scan and add a description for reference.

4. Configuring Scan Settings

- In the 'Targets' field, enter the IP addresses or hostnames of the systems you want to scan. Make sure you have authorization to scan these targets.
- You can specify advanced settings like scan schedules, notifications, and other preferences depending on the type of scan you have chosen.
- If necessary, you can also configure authentication settings to scan internal elements of the target system.

5. Starting the Scan

- After configuring all the settings, click on the 'Save' button to save your scan settings.
- To initiate the scan, click on the 'Launch' button.

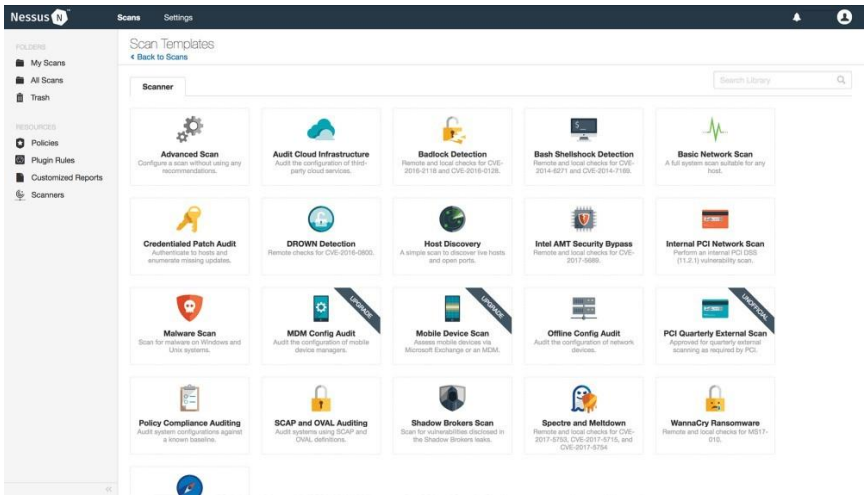
6. Monitoring the Scan

- Once the scan is initiated, you can monitor its progress in the dashboard.
- Nessus will display various metrics and statistics related to the scan, like the number of vulnerabilities found, the severity levels, etc.

7. Analyzing the Results

- After the scan is completed, click on it to view the detailed report.
- The report will categorize vulnerabilities as Critical, High, Medium, Low, or Info based on their severity.
- You can click on each vulnerability to get more details, like a description of the issue, the impact, and recommended solutions.

Output:



Result:

Thus, the Scanning a Target Using Nessus has been successfully done and executed.

Ex:No.: 7	View and capture network traffic using Wireshark.

Aim:

To View and capture network traffic using Wireshark.

Procedure:

1. Download Wireshark:

- Visit the official Wireshark download page at https://www.wireshark.org/download.html.
- Download the appropriate version for your operating system (Windows, macOS, or Linux).

2. Install Wireshark:

- Run the downloaded installer.
- Follow the installation wizard's instructions.

3. Launch Wireshark:

- Once the installation is complete, open the Wireshark application.

4. Select Network Interface:

- Wireshark will display a list of available network interfaces.
- Choose the network interface through which you want to capture traffic (e.g., Ethernet or Wi-Fi).

5. Start Capturing Traffic:

- **Click on the "Start" or "Capture" button to begin capturing network traffic.**

6. Analyze Captured Packets:

- While capturing, you can analyze the live stream of network packets.
- Use filters and search features in Wireshark to focus on specific types of packets or protocols.

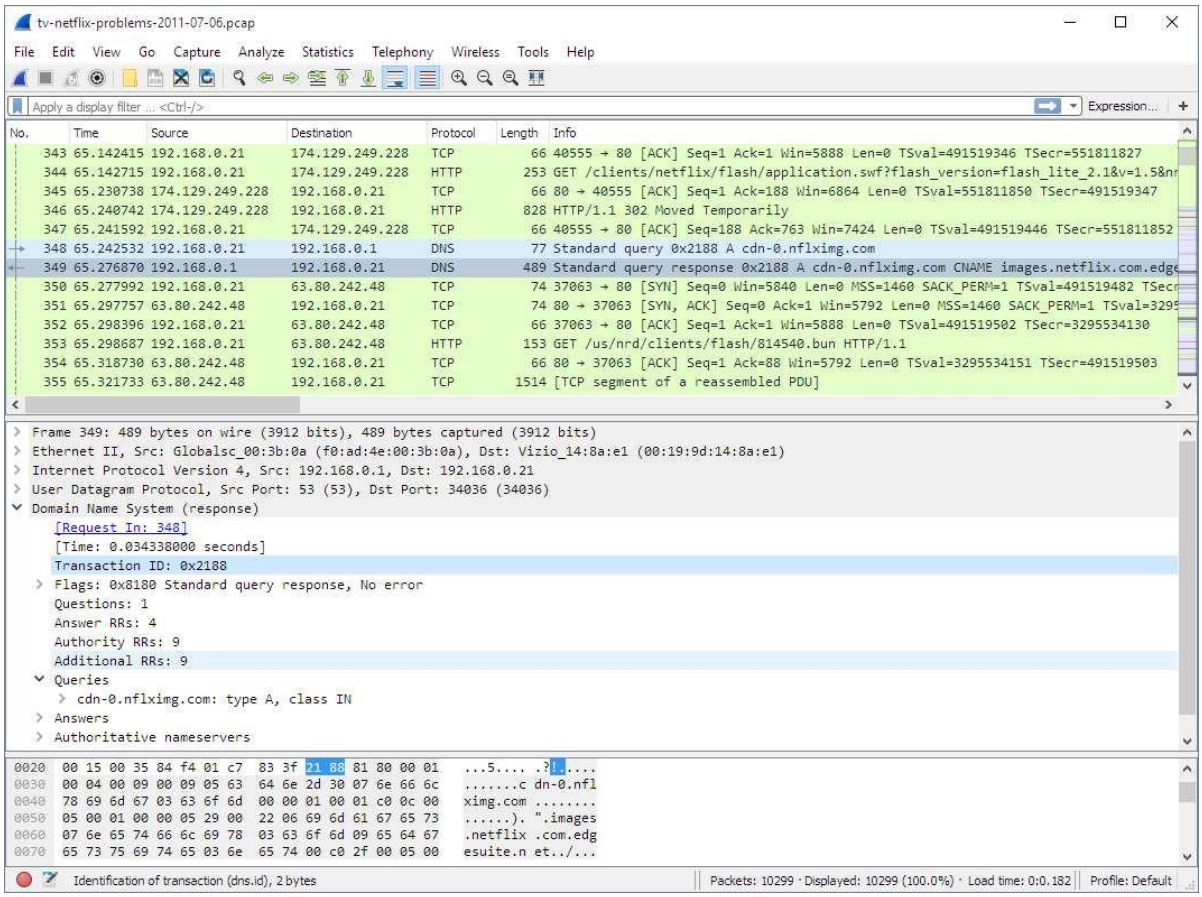
7. Stop Capturing:

- When you have captured enough data or want to stop the capture, click the "Stop" or "Capture" button again.

8. View Captured Packets:

- After stopping the capture, you can view the captured packets.
- Click on any packet to see its details, including source and destination IP addresses, protocols, and packet content.

Output:



Result:

Thus, Viewing and capturing network traffic using Wireshark has been successfully done and executed

Ex:No.: 8	**Automate dig for vulnerabilities and match exploits using Armitage FOCA**

Aim:

To Automate dig for vulnerabilities and match exploits using Armitage FOCA

Procedure:

1. Scanning and Reconnaissance:

- Launch Kali Linux and open Armitage.
- Perform network scanning using tools like Nmap to identify open ports and active hosts.
- Utilize Armitage to import the scan results and create a target list.

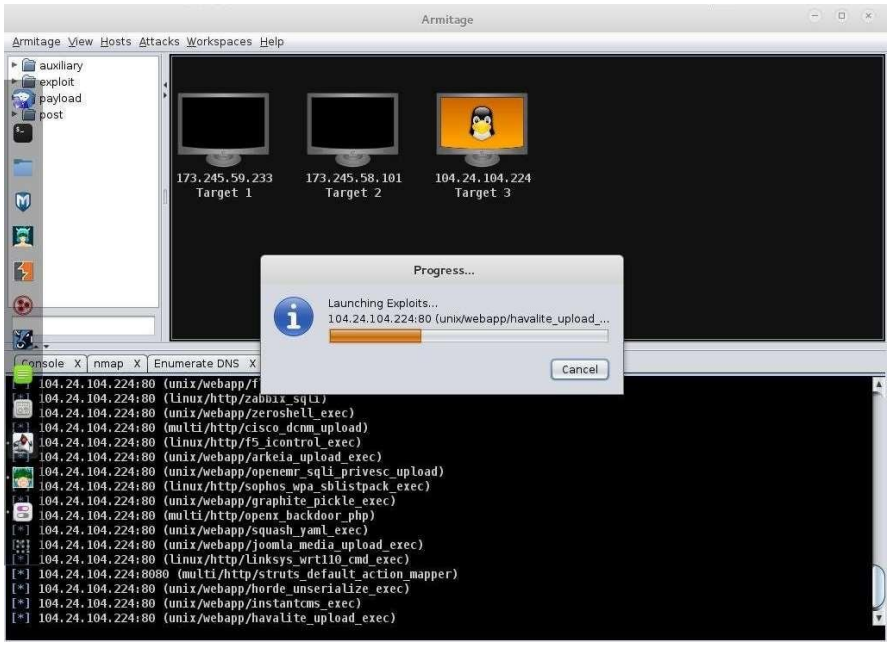
2. Vulnerability Analysis:

- Conduct vulnerability scanning on the target hosts using Armitage's built-in features.
- Identify potential vulnerabilities in the scanned systems, such as outdated software, weak passwords, or misconfigurations.
- Use FOCA (Fingerprinting Organizations with Collected Archives) to gather information about the target organization's documents and metadata.

3. Exploit and Attack:

- In Armitage, explore available exploits and payloads for the identified vulnerabilities.
- Select an appropriate exploit and payload combination for the target system.
- Launch the attack on the target system.

Output:



Result:

Thus, Automate dig for vulnerabilities and match exploits using Armitage FOCA has been successfully done.